3. cvičení

Příklady

IES FSV UK · Josef Hlahulek

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79. $\displaystyle\lim_{n\to\infty} \frac{\left(1+\frac{1}{n}\right)^{120} - \left(1+\frac{2}{n}\right)^{80}}{\left(1-\frac{1}{n}\right)^{100} + \left(1+\frac{3}{n}\right)^{100} - 2}$

85. $\displaystyle\lim_{n\to\infty} \frac{\left(1+\frac{1}{n}\right)^{150} - \left(1+\frac{2}{n}\right)^{60}}{\left(1-\frac{1}{n}\right)^{99} + \left(1+\frac{3}{n}\right)^{99} - 2}$

89. $\displaystyle\lim_{n\to\infty} \frac{\sqrt[3]{n^3+5n} - \sqrt[3]{n^3+2}}{\sqrt{n^4+n+1} - \sqrt{n^4+1}}$

98. $\displaystyle\lim_{n\to\infty} \frac{\sqrt[3]{n^3+n+1} - \sqrt[3]{n^3-n+2}}{\sqrt{n^4+2n+3} - \sqrt{n^4+n+4}}$

115. $\displaystyle\lim_{n\to\infty} \frac{(n+7)^{50} - (n^2+1)^{25}}{\sqrt{n^{100}+n^{99}-1} - \sqrt{n^{100}+2n^{99}+1}}$

48. $\displaystyle\lim_{n\to\infty} \frac{\sqrt[3]{n^3+6n} - \sqrt[3]{n^3+7}}{\sqrt{n^2+4} - \sqrt{n^2+1}}$

50. $\displaystyle\lim_{n\to\infty} \frac{\sqrt[n]{2^n+3^n}}{\sqrt[2n]{4^n+\sqrt{n}}}$

65. $\displaystyle\lim_{n\to\infty} \sqrt[n]{n^2+n^3+n^4+2^n+3^n+4^n}$